

THE BUBBLE FIELD COMPRESSION BOUNDARY CLOSURE

What the field can recover, what needs a constitutive key, and where the compression must stop

Joseph Shields

Formal reconstruction and reproducibility with ChatGPT

Core rule

Compression Sufficiency Rule The field is required to recover every quantity that is invariant under its compression map. It is not required - and is mathematically unable - to recover distinctions erased by that map unless an additional constitutive or measurement key is supplied.

This edition supersedes the optimistic weak-scale and lepton grading in the previous Closure Attack. The lepton ansatz is withdrawn; the electroweak coefficient is reclassified as matching; the exact source dictionary and compression theorem are retained.

Executive closure

The exact stopping point is now known. In the local scalar-phase sector, four cell variables are compressed into three field variables. The map has rank three and a one-dimensional kernel. Therefore the Universal Field of Least Action can reconstruct the complete metric/source/phase description, but metric observations cannot split light-barrier deformation from number-density deformation.

$$C(\alpha_M, \alpha_L, v, \theta) = (A, B, \Theta) = (-\alpha_M, \alpha_L + v/3, \theta)$$

$$\text{rank}(\text{DC})=3, \quad \text{ker}(\text{DC})=\text{span}\{(0,1,-3,0)\}$$

one exact hidden local degree

Compression rank	Hidden split	G candidate	UV/IR bridge
3	1	6.6643853e-11	5.242
of 4 local variables	α_L versus v	$\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	TeV

What “closed out” means here Every claim is assigned to its mathematically correct class: direct compressed result, conditional constitutive result, calibrated matching, algebraic translator, outside-resolution target, or rejected claim. An outside-resolution quantity is not counted as a failure of the compression; a missed quantity inside the declared output space is.

Claim	Field location	Final status	Closure statement
Golden fixed point, channel weights, traversal floor	INTERNAL INVARIANT	RESULT	Exact algebra once the physical golden realization is admitted.
Cell scale $a_0(W)=(W \hbar c/\rho_Lambda)^{(1/4)}$	DIMENSIONAL OUTPUT	CONDITIONAL RESULT	Exact family; the branch $W=1$ or $W=W_M$ is a declared vacuum-channel assignment.
$w=-1$	MACRO INVARIANT	CONDITIONAL RESULT	Follows exactly from the continuity equation if cell renewal keeps ρ_Lambda constant during expansion.
Metric/source sector in A and B	COMPRESSED FIELD	EXACT REPRESENTATION	GR determines A and B; all metric observables factor through them.
Separate α_L and ν	KERNEL DEGREE	NOT IDENTIFIABLE FROM METRIC	Requires a nonmetric observation or a constitutive section.
$\alpha_L/\alpha_M = r/u = 1/\sqrt{5}$	CONSTITUTIVE SECTION	CONDITIONAL THEOREM	Forced by least Euclidean response cost under the mixed-boundary constraint $r \alpha_L + u \alpha_M = \chi$.
Schwarzschild, interior star, factor-two lensing	COMPRESSED FIELD	EXACT GR RECONSTRUCTION	Exact in the spherical scalar sector; not an independent derivation of GR from cell microdynamics.
G from $q=r^{241}(1-r^3)^2$	DISCRETE BRIDGE	POSTDICTIVE CONDITIONAL	Conditional theorem under 240-

Claim	Field location	Final status	Closure statement
		FORECAST	mode factorization, one readout, and two retained interrogative legs; frozen for future Lambda tests.
UV/IR scale $E_{\text{cross}}=\sqrt{E_P E_0}$	STRUCTURAL SCALE	RESULT/IDENTITY	Exact energy form of $a_0=\sqrt{l_P L_{\text{Lambda}}}$ after the G bridge is supplied.
Electroweak VEV 246 GeV	LOW-ENERGY MATCHING	CALIBRATION, NOT DERIVATION	The field outputs E_{cross} ; the coefficient v/E_{cross} is a matching datum until a symmetry-breaking operator fixes it.
Charged-lepton masses	ERASED SPECTRAL DETAIL	PREVIOUS ANSATZ REJECTED	Exact masses require a Yukawa/mass operator; the prior e, mu, tau formulas are withdrawn.
QCD string tension	DIMENSIONAL TRANSLATOR	IDENTITY, NOT PREDICTION	$\sigma=E^2/(\hbar c)$ converts a supplied tension to a supplied local scale; it does not select the QCD scale.
Vacuum crowding energy as dark matter	CLAIMED MACRO OUTPUT	EXCLUDED	Short by about 11 orders in the existing audit; geometric crowding is not the same claim as extra energy density.
Range hierarchy	OPERATOR CLASS	CONDITIONAL RESULT	Green-function ranges follow exactly after kinetic/mass operator assignments; channel weights alone do not force those assignments.

1. The exact compression boundary

The field is a compression algorithm in the precise mathematical sense: it maps a richer local state to a smaller set of invariant readout variables. The right question is therefore not “does the compressed description reproduce every microscopic label?” but “which functions survive the quotient?”

1.1 Compression Factorization Theorem

Theorem Let $C:X \rightarrow Y$ be the field compression and $Q:X \rightarrow Z$ a proposed output. Q is recoverable from the compressed field if and only if Q is constant on every fiber $C^{-1}(y)$. Equivalently, there exists a unique reduced function $\tilde{Q}$ on the image of C such that $Q = \tilde{Q} \circ C$. For differentiable maps, dQ must annihilate every vector in $\ker(DC)$.

$$Q \text{ recoverable} \Leftrightarrow C(x)=C(x') \text{ implies } Q(x)=Q(x') \Leftrightarrow Q=\tilde{Q} \circ C$$

$$\text{local test: } \ker(DC_x) \subseteq \ker(dQ_x)$$

1.2 The Bubble Field map and its exact null direction

Take the full local state to be the matter-clock deformation α_M , light-barrier deformation α_L , number-density deformation v , and phase θ . The compressed field retains the clock potential A , total spatial packing B , and phase Θ .

$$q=(\alpha_M, \alpha_L, v, \theta), \quad y=(A, B, \Theta)=(-\alpha_M, \alpha_L + v/3, \theta)$$

$$DC = [[-1, 0, 0, 0], [0, 1, 1/3, 0], [0, 0, 0, 1]]$$

The unique local null direction is

$$\delta\alpha_M=0, \quad \delta\alpha_L=\delta, \quad \delta v=-3\delta, \quad \delta\theta=0$$

Moving along this direction changes the proposed microscopic decomposition but leaves A , B , Θ , the metric, curvature, geodesics, clocks, lensing and phase readout unchanged. That is the exact information budget of the current scalar-phase field.

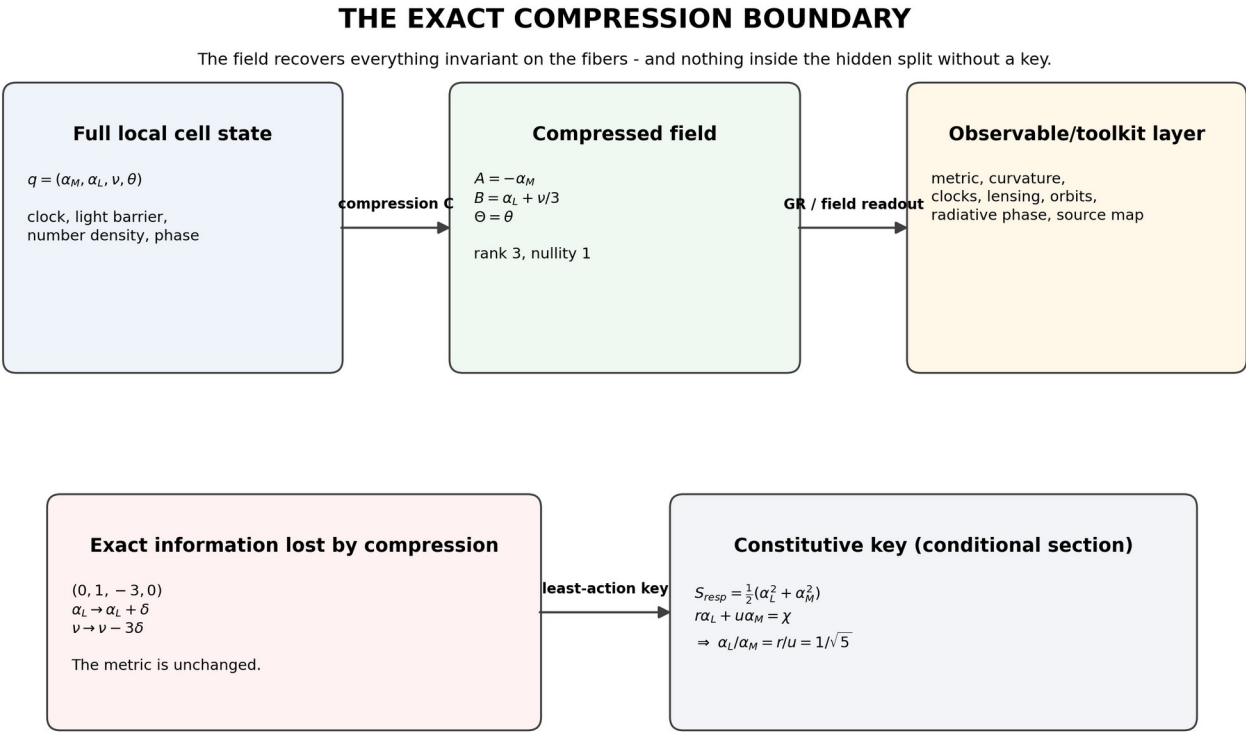


Figure 1. The exact compression boundary. The field retains three local combinations and erases one light-density split. A constitutive key selects one representative from each fiber.

1.3 Bottom-up and top-down meet at the compressed field

Bottom-up derivation and top-down observation do not have to reconstruct the same microscopic story. They have to meet on the same compressed invariants:

microphysical rules → (A,B,Θ) ← observations and established toolkits

- Bottom-up: the fixed-point algebra, cell rules and constitutive laws calculate the compressed field.
- Top-down: clocks, orbits, lensing, redshift and radiation reconstruct the same field variables.
- Validation occurs in the overlap. A unique microscopic split is not implied unless the experiment couples to the hidden direction or a constitutive law selects it.

No escape clause Compression does not excuse a failed prediction. If a quantity is a function of A, B or Θ and the field predicts it incorrectly, the claim is falsified. The threshold only prevents demands for information that the map explicitly erased.

1.4 The five gates for every future claim

Gate	Question	Failure status
Fiber gate	Is the target invariant on every compression fiber?	If no: outside current resolution.
Anchor gate	Can its physical units be built from	If no: dimensional scale remains open.

	declared measured anchors?	
Bridge gate	Is every constitutive or representation map explicit?	If no: conditional/open bridge.
History gate	Was the expression fixed before viewing the target?	If no: postdiction or calibration.
Precision gate	Does an in-scope prediction survive the measurement uncertainty?	If no: rejected, regardless of percentage closeness.

2. Observation, calibration, and prediction are different jobs

The same measured number may enter a calculation in different roles. The paper must name the role before the calculation, because confusing an input with a prediction is the main source of false closure.

Role	Meaning	Allowed claim
Anchor	Measured quantity supplies units or a boundary condition.	Conditional dimensional result.
Calibration	Measured target fixes a matching coefficient.	Translator/model calibration, not derivation.
Validation target	Target is not used to construct the formula.	Independent test of an existing result.
Frozen forecast	A postdictive formula is locked before new data arrive.	Future evidence can support or reject it.

Two-axis grading Every statement now receives two grades. Derivational status asks whether it follows from the declared model. Evidential status asks whether it was predicted, calibrated, or chosen after seeing the observation. A formula can be a conditional theorem inside the model and still be a postdiction experimentally.

3. Source-sector closure and the constitutive key

General relativity is used as the macroscopic toolkit. The Bubble Field is the compressed decoding of the metric, not an extra minimally coupled scalar stress tensor in the exterior.

$$ds^2=-e^{\{2A(\rho)\}}c^2dt^2+e^{\{2B(\rho)\}}(d\rho^2+\rho^2d\Omega^2)$$

$$A=-\alpha_M, \quad B=\alpha_{L+v/3}$$

$$G_{tt}=-e^{\{-2B\}}(B'^2+2B''+4B'/\rho)$$

$$G_{rr}=e^{\{-2B\}}(2A'B'+B'^2+2(A'+B')/\rho)$$

$$G_{\theta\theta}=e^{\{-2B\}}(A'^2+A''+B''+(A'+B')/\rho)$$

Macroscopic result Given a relativistic energy density and pressure profile, the Einstein equations determine A and B. This completes the GR-to-bubble source dictionary in the isotropic spherical sector. It

does not yet derive arbitrary matter stress-energy from microscopic cells.

3.1 Exact identifiability theorem

The metric determines $\alpha_M = -A$ exactly and determines only the combination $\alpha_L + v/3 = B$. No metric-only experiment can identify the two pieces separately, at any precision. A nonmetric coupling or a constitutive section is required.

$$(\alpha_L, v) \sim (\alpha_L + \delta, v - 3\delta)$$

3.2 Least-action constitutive section

The smallest explicit closure uses the mixed boundary covector (r, u) and an isotropic quadratic response cost. For a required boundary response χ , extremize

$$S_{\text{resp}} = \frac{1}{2}(\alpha_L^2 + \alpha_M^2) - \lambda(r\alpha_L + u\alpha_M - \chi)$$

Stationarity gives

$$\alpha_L = \lambda r, \quad \alpha_M = \lambda u, \quad \alpha_L / \alpha_M = r/u = 1/\sqrt{5}$$

Status of the ratio The ratio is no longer an unexplained numerical preference. It is a conditional theorem of the declared least-cost response action. The remaining physical question is whether the microscopic boundary response is in fact measured by this Euclidean cost and mixed covector.

On the weak Schwarzschild branch $B \approx \alpha_M$, the density response is therefore

$$v \approx 3(1 - 1/\sqrt{5})\alpha_M = 1.6583592135 \alpha_M$$

weak-field positive crowding

In the exact isotropic solution, use $v = 3[B(\alpha_M) - (1/\sqrt{5})\alpha_M]$; the weak linear coefficient is not substituted into the strong field.

3.3 Exact exterior, interior and light bending

$$A = \ln[(1-x)/(1+x)], \quad B = 2\ln(1+x), \quad x = GM/(2\rho c^2)$$

$$B(\alpha_M) = \alpha_M - 2\ln \cosh(\alpha_M/2)$$

The symbolic vacuum Einstein residuals vanish. The constant-density star reconstructs end to end; on the constitutive section the example with compactness 0.30 has central $n/n_0 = 1.4766$. These are exact GR reconstructions in bubble coordinates.

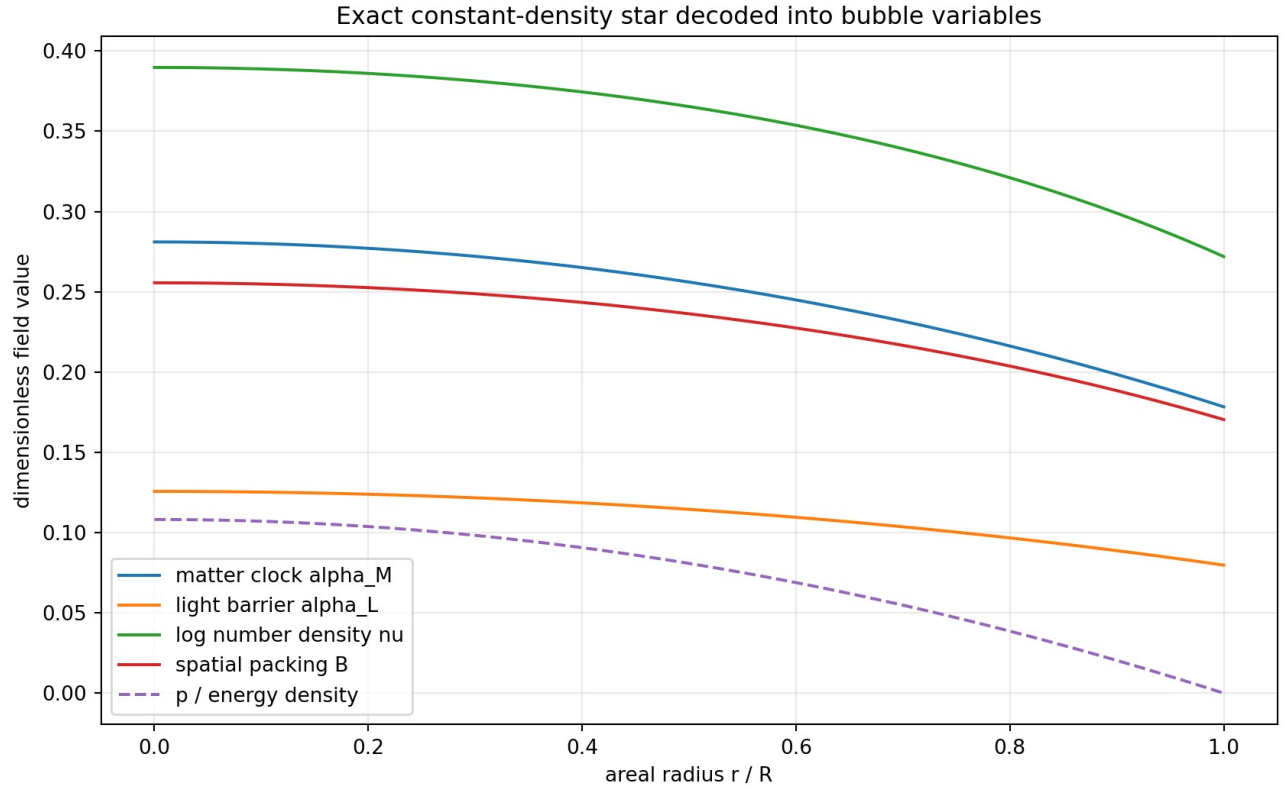


Figure 2. Constant-density relativistic star decoded into the clock, light-barrier, density and total packing variables on the least-action constitutive section.

The familiar factor of two in weak light deflection is recovered because slow motion reads the clock potential while lensing reads both clock and spatial packing. With $\gamma=1$, the spatial mode supplies the second half. This is an exact consequence of the GR dictionary, not an independent derivation of GR.

4. Cell scale, vacuum equation of state, and the dimensional boundary

4.1 The cell is an exact family, not one unqualified number

For a causal cell with $\varepsilon\tau=\hbar$, $a=c\tau$, baseline number density $n_0=a_0^{-3}$, and a channel fraction W carrying the observed vacuum energy density,

$$a_0(W)=(W\hbar c/\rho_\Lambda)^{1/4}$$

Vacuum assignment	W	Cell scale
Whole cell	1	87.751 μm
Matter channel	r^2	48.780 μm

The formula is the result. Choosing which channel the observed $\rho\Lambda$ represents is the physical branch. The paper must never quote 87.8 μm and 48.8 μm as if both were unconditional predictions of the same assignment.

4.2 Exactly when $w=-1$ follows

The cosmological continuity equation is

$$d(\rho V)=-p\,dV$$

If cell renewal during expansion keeps the vacuum energy density constant, $d\rho=0$, then

$$\rho\,dV=-p\,dV \quad \Rightarrow \quad p=-\rho \quad \Rightarrow \quad w=-1$$

Compression threshold for w Fixed action per cell does not by itself prove constant density. The result becomes exact after the cell-renewal/non-dilution rule is declared. The field can compress that rule into $w=-1$; it cannot infer the rule from w alone.

4.3 G and the cosmological hierarchy are one dimensionless bridge

$$q=\Lambda\ell_P^2/(8\pi)=\Lambda\hbar G/(8\pi c^3)$$

$$a_0=\sqrt{(\ell_P L_\Lambda)}, \quad L_\Lambda=\sqrt{(8\pi/\Lambda)}$$

These identities show that the Newton coupling, the vacuum hierarchy and the UV/IR cell separation encode the same missing dimensionless normalization. This is an exact reframing even before a value of q is supplied.

5. The G bridge: conditional theorem, postdictive evidence, frozen test

The registered bridge is

$$q_{UF}=r^{241}(1-r^3)^2$$

$$G_{UF}=[8\pi c^3/(\hbar\Lambda)]\,r^{241}(1-r^3)^2$$

Inside the declared representation, the factors have distinct jobs: 240 internal root modes, one external readout, and two retained legs of the emit-return interrogative cycle. If those three physical identifications hold and multiply independently, the formula follows without a continuous parameter.

Derivational grade Conditional theorem under the 240-mode factorization, external-readout and two-leg retention postulates.

Evidential grade Postdiction: the two-leg prefactor was identified after the target hierarchy was known. The expression is now frozen. Future improvement in the geometric Λ boundary is the clean test; no alternate power or retention factor may be substituted afterward.

Quantity	Value	Role
q_{UF}	1.14744854e-123	frozen model output
G_{UF}	6.66438529e-11 m ³ kg ⁻¹ s ⁻²	conditional forecast from Λ
Difference from comparison G	-0.149%	current postdictive residual

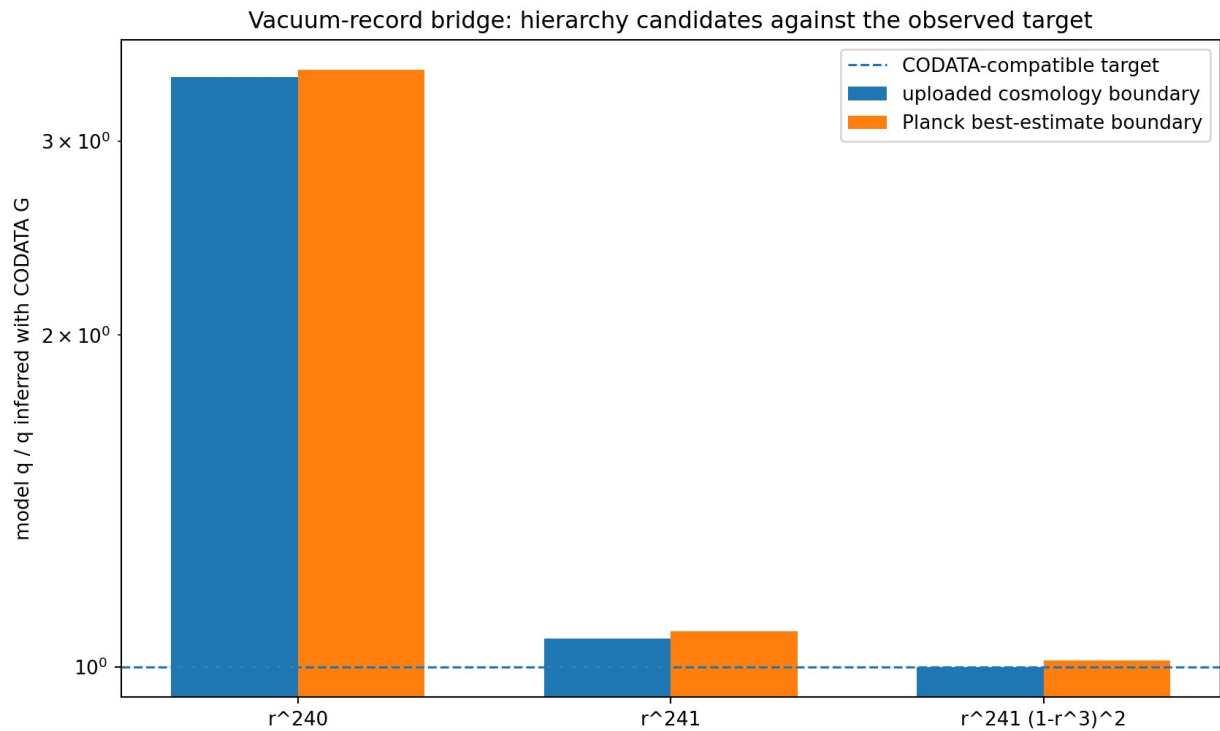


Figure 3. Discrete vacuum-record candidates. The two-leg formula is frozen as the future cosmological test branch.

6. The exact scale output and where low-energy matching begins

6.1 Structural UV/IR scale

$$E_0 = \hbar c / a_0, \quad E_P = \hbar c / \ell_P, \quad E_\times = \sqrt{(E_P E_0)}$$

On the reproducibility branch $E_\times = 5.2416$ TeV. This is the energy form of the exact geometric-mean identity $a_0 = \sqrt{(\ell_P L_\Lambda)}$. It is the last scale supplied by the current compression without an additional low-energy operator.

6.2 Electroweak VEV is a matching problem

$$v = c_{EW} E_\times$$

The observed matching coefficient is $c_{EW} = v/E_\times = 0.046973905$. The earlier expression $(r^2/2)\sqrt{(1-r^3)} = 0.047036024$ was chosen because it landed near v ; it is not retained as a derivation.

Exact stopping point The compression produces the crossover scale $E_\times$. A Higgs or symmetry-breaking operator must determine c_{EW} , the gauge couplings and the weak mixing angle. Until that operator exists, using the measured v is calibration - not a failure of the compression and not a prediction.

6.3 Charged leptons are beyond the current resolution

In the Standard Model toolkit,

$$M_\ell = (v/\sqrt{2}) Y_\ell$$

The Bubble Field currently supplies neither the Yukawa operator Y_ℓ nor an eigenvalue theorem forcing generation depths. Therefore exact electron, muon and tau masses vary inside unresolved spectral detail and do not pass the Fiber/Bridge gates.

Withdrawal The previous formulas $m_e = v r^{11}(1-r/2)$, $m_\mu = m_e \phi^{11}(1+r^3)$, and $m_\tau = m_e \phi^{17}(1-r^3)$ are removed from the active theory. Their sub-percent proximity is not enough at the precision of lepton data. They remain only as a documented rejected ansatz.

A future lepton claim enters scope only after the theory constructs a localized phase/winding operator, proves its spectrum and fixes the dimensional matching without using the lepton masses as targets.

7. Translators, no-go results, and operator-dependent outputs

7.1 QCD tension is a translator identity

$$\sigma=\rho a^2=E/a=E^2/(\hbar c), \quad a=\hbar c/E$$

Supplying σ and recovering E is algebraic inversion. The field has not selected the QCD scale. The vacuum-cell identification remains excluded by roughly 10^{23} in the existing audit.

7.2 Dark matter: distinguish energy from geometry

The literal claim that vacuum-cell crowding energy supplies the missing galactic dark-matter energy density is excluded by roughly eleven orders. That no-go does not automatically exclude a relational or geometric crowding interpretation, because metric reconstruction and extra material energy are different observables. Any revived dark-sector claim must state which one it means.

7.3 Range hierarchy begins at the operator assignment

Mode	Declared operator	Green function	Status
Light	$-\nabla^2$	$1/r$	infinite after massless kinetic assignment
Boundary	$-\nabla^2+m_B^2$	$e^{-m_B r}/r$	finite after massive kinetic assignment
Matter	m_M^2 without spatial kinetic term	local algebraic response	zero/contact after auxiliary assignment

The ranges follow exactly from these operators. The three channel weights and block counts alone do not uniquely force the operators, so the range hierarchy remains a conditional representation result.

8. The field recovery threshold - final statement

The Universal Field of Least Action is neither required nor permitted to claim unlimited bottom-up reconstruction. Its formal scope is the set of observables that factor through its compression map, plus quantities obtained after explicitly declared anchors and constitutive sections.

Recoverable set $R_C = \{Q \mid Q = \tilde{Q} \circ C\}$

Conditional set $R_{\{C,s\}} = \{Q \mid Q = \tilde{Q} \circ s \circ C \text{ for a declared section } s\}$

Outside set $O_C = \{Q \mid Q \text{ changes along } \ker(DC) \text{ and no extra channel is supplied}\}$

Operational threshold At the present scalar-phase level, the field gets A , B and Θ and every invariant made from them. It gets α_M from A . It does not get α_L and v separately without the constitutive key. It gets the structural UV/IR scale, but not low-energy spectral eigenvalues without a matching operator. That is the exact point at which the inference stops.

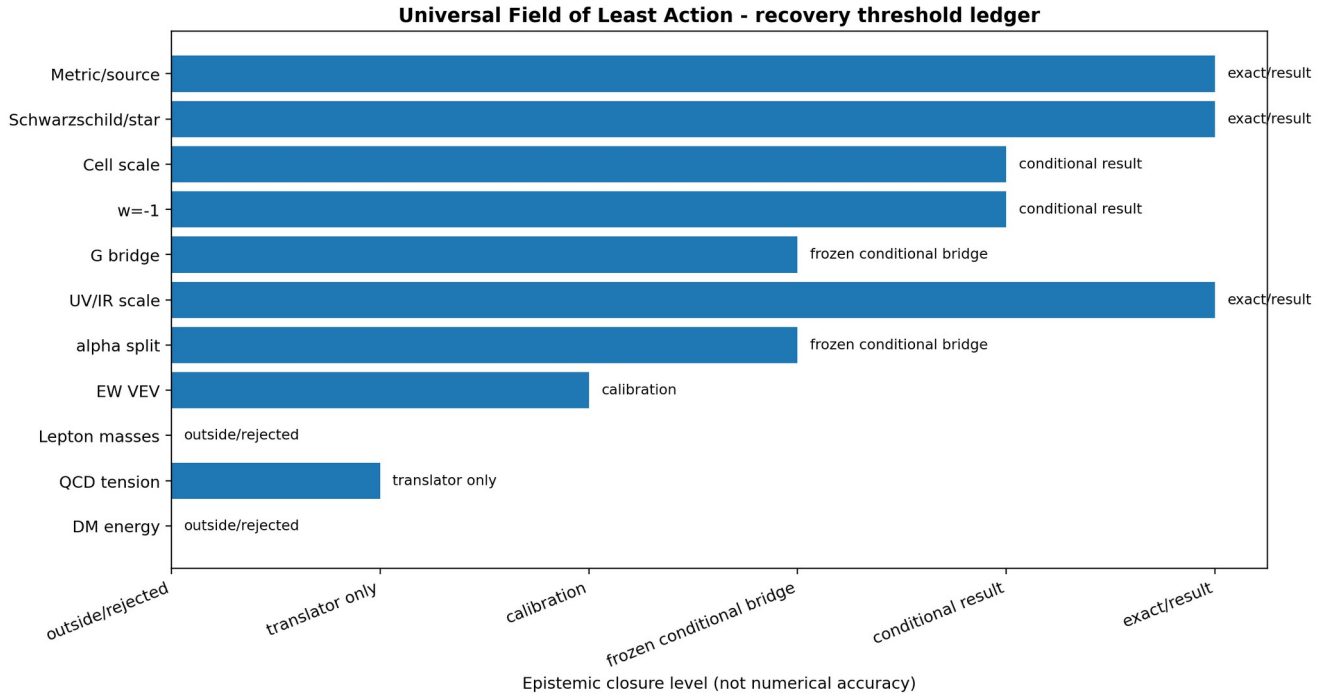


Figure 4. Final recovery ledger. The horizontal position describes derivational closure, not numerical closeness.

8.1 The claim-writing protocol

1. Write the compression map and its kernel for the sector being studied.
2. Name all measured quantities as anchors, calibrations, validation targets or frozen forecasts.
3. Test fiber invariance before attempting a numerical formula.
4. Add a constitutive section only through an explicit action or response law.
5. Do not label a translator identity as a prediction.

6. Reject an in-scope precision formula when it misses, even if its percentage error appears small.
7. Do not call an out-of-scope microscopic detail a failure of the compressed field; call it an operator still absent from the representation.

8.2 What is closed now

- The local information budget: rank three, nullity one.
- The exact metric/source dictionary in the isotropic spherical sector.
- The exact reason gravity cannot identify the light-density split.
- A smallest conditional least-action section giving $\alpha_L/\alpha_M=1/\sqrt{5}$.
- The exact cell-scale family and conditional $w=-1$ theorem.
- The exact G/Λ hierarchy identity and one frozen conditional G bridge.
- The exact UV/IR crossover scale and the precise point where electroweak matching begins.
- The withdrawal of the charged-lepton ansatz and QCD “prediction” language.
- The exclusion of vacuum crowding energy as galactic dark matter.

8.3 What remains genuinely physical

- Derive the 240-mode vacuum projector and external readout as a physical stress-energy observable, rather than a representation postulate.
- Derive or falsify the Euclidean mixed-boundary response action microscopically.
- Extend the compression map beyond the spherical scalar sector to shear, rotation and general tensor geometry.
- Construct the symmetry-breaking and mass operators that map $E_\times$ to Standard Model couplings and spectra.
- Derive an observable short-distance transfer function for the cell scale; a length by itself is not a laboratory signal.

Appendix A. Reproducibility certificate

The accompanying compression-boundary script symbolically calculates the compression rank and kernel, tests representative recoverability directions, derives the least-action constitutive ratio, freezes the G bridge, calculates the scale-matching boundary, and writes the final claim ledger.

File	Role
bubble_field_compression_boundary.py	compression theorem, constitutive derivation and status audit
bubble_field_compression_boundary_results.json	machine-readable certificate
bubble_field_compression_boundary_map.png	exact compression diagram
bubble_field_recovery_ledger.png	claim-threshold visualization
bubble_field_closure_attack.py	source equations, star reconstruction and G calculation
bubble_field_closure_attack_results.json	prior numerical certificate, regraded here

Appendix B. Status vocabulary

Term	Meaning
Exact result	Follows from the declared equations and lies inside the compressed output space.
Exact representation	An established toolkit solution is reconstructed without residual in the field variables; this is not automatically a microscopic derivation of the toolkit.
Conditional theorem	Follows exactly after named physical postulates or a constitutive section are admitted.
Calibration	An observed target fixes a matching coefficient.
Frozen forecast	A postdictive formula is locked and exposed to genuinely new data.
Translator	An identity converts one supplied physical quantity into another supplied scale.
Outside resolution	The target varies along a compression kernel or requires an absent operator.
Rejected	A declared in-scope formula fails empirical precision or a structural no-go.

Appendix C. Source lineage

8. Shields, J. The Bubble Field: Gravity as overcrowding in a sea of least-action cells. Working paper, 29 July 2026.
9. Shields, J. Least-Cost Holonomy and Co-Distinction Fields. Working paper, July 2026.
10. Shields, J. Unified Mechanics bridge-closure corpus. Golden partition, traversal floor, conditional E8 vacuum projector and external readout.
11. Exact toolkit solutions: isotropic Schwarzschild exterior and constant-density Schwarzschild interior.
12. Planck 2018 cosmological boundary and 2022 CODATA comparison values as recorded in the prior reproducibility package.

END OF COMPRESSION BOUNDARY CLOSURE
